Research and teaching prototype. Outputs are suspected tumour regions predicted by the model and estimated volumes. Not a diagnosis; not clinically validated; not for clinical use.

NeuroScope 3D — estimated tumour-region volumes

Study brats_d6fff866ad9e7f57cfde022984eda4dc · generated 2026-10-16T21:06:06+00:00

Input

Sequence	File	SHA-256
t1gd	T1Gd_BRATS_002_t1ce.nii.gz	19753d7b6f90b4b37a6df89d6381d6b17408439e6feec31e726da7cb36499471
t1	T1_BRATS_002_t1.nii.gz	8974b0b277957b6f267bb220e18e2eeac04bfdf0d14503e67e64781eb7a39087
t2	T2_BRATS_002_t2.nii.gz	8f85bd990b11f2d48522e1e50e07dbc187badb728da2d6c8d39245582e74cf5e
flair	FLAIR_BRATS_002_flair.nii.gz	ffc10ff02fa68a60cf3cda2d72a75480d120531e9393bdc965c407dbe81e9168

Grid [240, 240, 155] voxels, spacing 1.000, 1.000, 1.000 mm, orientation RAS. Expert mask: supplied.

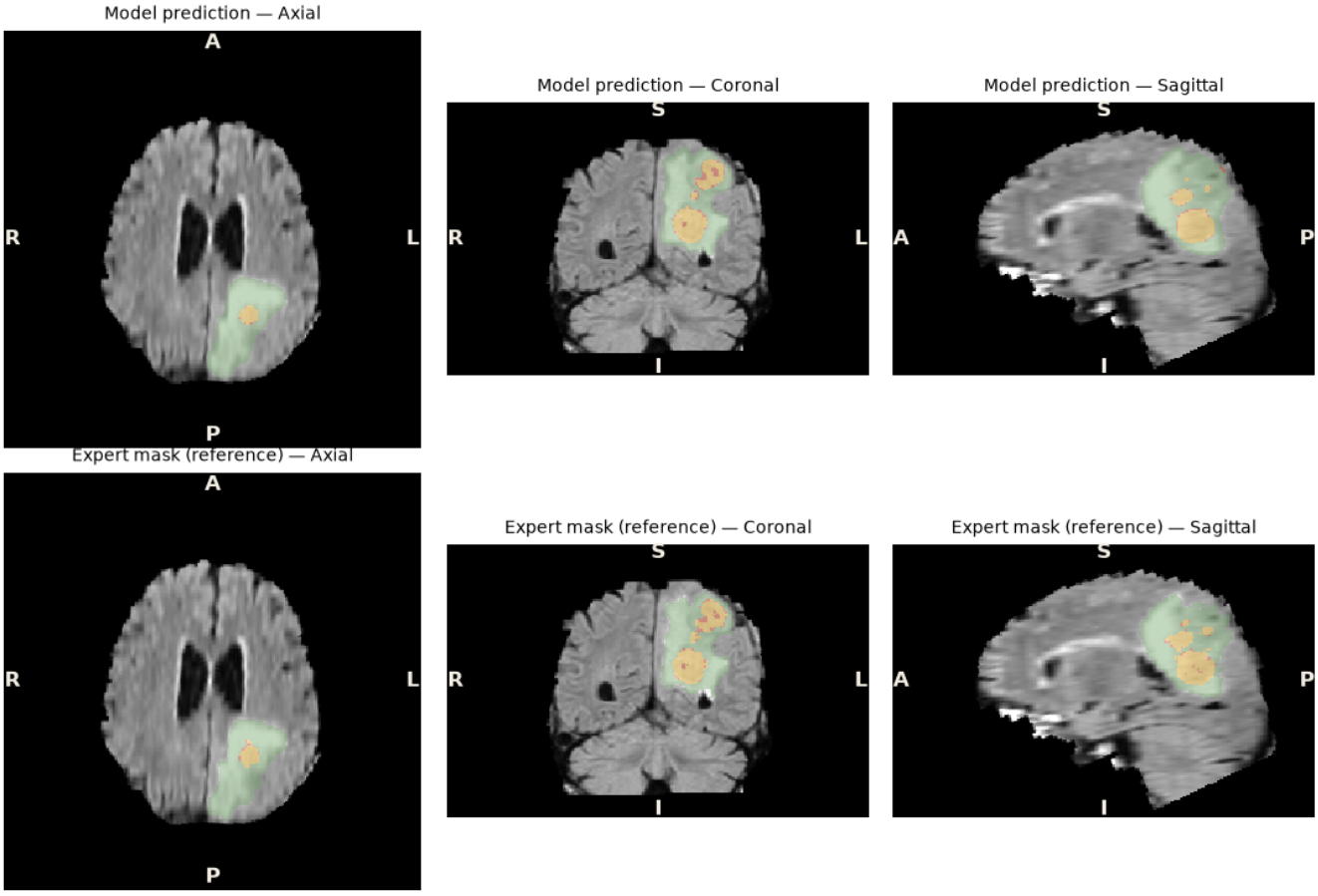
Suitability warnings: none.

Estimated volumes of suspected regions predicted by the model

Formula: $V = N \times (\Delta x \times \Delta y \times \Delta z) = N \times (1.000 \times 1.000 \times 1.000)$ mm³ = N × 1.000 mm³; mL = mm³ / 1000 (1 mL = 1,000 mm³).

Region	Model voxels N	Model mm³	Model mL	Expert voxels	Expert mL
NCR (label 1)	2118	2,118.0	2.12	1106	1.11
ED (label 2)	58036	58,036.0	58.04	53736	53.74
ET (label 4)	7584	7,584.0	7.58	8219	8.22
TC (labels 1+4)	9702	9,702.0	9.70	9325	9.32
WT (labels 1+2+4)	67738	67,738.0	67.74	63061	63.06

FLAIR with overlay: red NCR (1), green ED (2), yellow ET (4). Canonical voxel (100, 70, 99). Suspected regions predicted by the model.



Agreement with the supplied expert mask (this case only)

Region	Dice	Sensitivity	HD95 (mm)
WT	0.920	0.954	2.8
TC	0.850	0.867	4.6
ET	0.873	0.839	2.8

How much to trust it

Test-time augmentation: the model was re-run on the scan mirrored along each axis (4 passes; the label map is from the un-augmented pass). 8.2% of WT voxels are disputed between passes; WT is 65.23 mL where all passes agree and 71.08 mL where any pass predicts it.

Region	Volume range across passes (mL)	Lowest Dice vs main pass
WT	67.21 – 69.47	0.977
TC	9.42 – 10.22	0.942
ET	7.55 – 7.75	0.967

Modality ablation — not clinical evidence. Each sequence in turn is replaced by zeros before inference; Dice compares the result with the full-input prediction. It shows what this model relies on, not what is in the patient.

Input change	WT Dice vs full	TC Dice vs full	ET Dice vs full
without T1GD	0.93	0.39	0.00
without T1	0.97	0.95	0.96
without T2	0.96	0.96	0.98
without FLAIR	0.59	0.87	0.93

Model evaluation (20 held-out labelled cases)

Medical Segmentation Decathlon Task01_BrainTumour (labelled training list), labels converted to BraTS codes. Cases: BRATS_013, BRATS_038, BRATS_048, BRATS_084, BRATS_086, BRATS_145, BRATS_169, BRATS_172, BRATS_175, BRATS_220, BRATS_300, BRATS_305, BRATS_315, BRATS_337, BRATS_346, BRATS_377, BRATS_395, BRATS_397, BRATS_413, BRATS_433.

Region	Dice mean ± SD	Sensitivity mean ± SD	HD95 median (mm)
WT	0.885 ± 0.108	0.932 ± 0.062	4.6
TC	0.851 ± 0.212	0.912 ± 0.092	2.2
ET	0.764 ± 0.274	0.857 ± 0.189	1.7

Overlap disclosure: We did not train or fine-tune the model, so none of these cases were used by us for training or tuning. However, the pretrained MONAI model was trained on BraTS 2018, and Decathlon Task01 is built from BraTS 2016 and 2017 cases, which BraTS 2018 largely re-uses. Decathlon renamed the cases, so we cannot tell which evaluation cases the model saw during its training; most probably overlap. These metrics are therefore likely optimistic and are not an independent held-out estimate.

Provenance

Item	Value
Model	MONAI brats_mri_segmentation (SegResNet), version 0.5.4
Licence	Apache-2.0 (MONAI Consortium)
Model training data	BraTS 2018 (285 cases; bundle authors' own split)
Preprocessing	Stack sequences as [T1Gd, T1, T2, FLAIR]; reorient to RAS; resample to 1 mm isotropic (bilinear); z-score intensity normalisation per channel over non-zero voxels; sliding-window inference with AMP; sigmoid; invert resampling/orientation back to the input grid (bilinear on probabilities); threshold 0.5; TC/WT/ET channels -> BraTS codes (ET=4, TC without ET=1, WT without TC=2). Uncertainty: 3 extra passes on the input mirrored along each axis (label map from the plain pass).
Code commit	f0429c60da636a45297181667c012bc1cbdf178a
Inference timestamp (UTC)	2026-10-16T21:05:38+00:00
Device / inference time	cuda roi=(240, 240, 160) / 4.81 s

Limitations

• Research and teaching prototype; not clinically validated and not approved for any clinical use. It does not provide diagnosis, grading, prognosis, treatment or surgical/radiotherapy planning.

• The pretrained model was trained by the MONAI team on BraTS 2018 (adult glioma, pre-operative, skull-stripped, co-registered, 1 mm). It may fail on other tumour types (metastases, meningioma, paediatric tumours), post-operative scans, non-skull-stripped or

non-co-registered images, unusual contrast, motion or other artefacts.

• Evaluation cases come from the Medical Segmentation Decathlon Task01, which derives from BraTS 2016/2017; most are likely included in the model's BraTS 2018 training data, so reported Dice/sensitivity are optimistic.

• Enhancing tumour (ET) and necrotic core (NCR) are small, low-contrast regions and are the least reliable; ET can be missed entirely or predicted where an expert would mark none (in our evaluation, non-enhancing tumours such as BRATS_175 and BRATS_169 received spurious ET/TC predictions).

• Images coarser than 1 mm are resampled to 1 mm; thick-slice data reduces accuracy and volume precision.

• Volumes are estimates (voxel count x voxel volume) of the model's predicted regions, not measurements of the tumour, and inherit all segmentation errors.

• Input suitability checks are simple heuristics (background fraction, tissue volume, mask agreement, size, shape, texture); passing them does not guarantee the scan is appropriate.

• The test-time-augmentation uncertainty only measures how the prediction changes when the scan is mirrored; a confidently wrong region can be unanimous across passes. Modality ablation is not clinical evidence.